

Code Generation

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Basic blocks

A basic block is a sequence of consecutive statements in which flow of control enters at the beginning and leaves at the end without halt or possibly of branching except at the end.

Example:

$$\begin{array}{ll} t_1 = a * a & t_2 = a * b \\ t_3 = 2 * t_2 & t_4 = t_1 + t_3 \\ t_5 = b * b & t_6 = t_4 + t_5 \end{array}$$

A three-address statement $x = y(z)$ is said to *define* x and to *use* or reference y and z .

A name in a basic block is said to be live at a given point if its value is used after that point in the program, perhaps in another basic block.

Algorithm

Partitioning a sequence of three-address statements into basic blocks.

Input: A sequence of three-address statements.

Output: A list of basic blocks with each three-address statement in exactly one block.

Method:

- a. Determine the set of leaders, the first statements of basic blocks. The rules we use are:
 - i. The first statement is the leader.
 - ii. Any statement that is the target of a conditional or unconditional goto is a leader.
 - iii. Any statement that immediately follows a goto or conditional statement is a leader.
- b. For each leader its basic block consists of the leader and all the statements upto but not including the next leader or the end of the program.

Example

Example 1: The following statements of source code. It computes the dot product of two vectors of length 20. A list of three-address statements performing this computation is also shown.

Source code:

```
begin
  prod = 0
  i = 1
  do begin
    prod = prod + a \* b
    i = i + 1
  end
  while i <= 20
end
```

Three-address code:

```

1. prod = 0 // leader by rule 1
2. i = 1
3. t_1 = a // leader by rule 2
4. t_2 = b
5. t_3 = t_1 + t_2
6. t_4 = prod + t_3
7. prod = t_4
8. t_5 = i + 1
9. i = t_5
10. if i <= 20 goto(3)

```

So, the basic blocks are:

1-2 3-10

Example 2:

```

1. i = m - 1 // leader by rule 1
2. j = n
3. t_1 = t * n
4. V = a[t_1]
5. i = i + 1 // leader by rule 2
6. t_2 = 4 * i
7. t_3 = a[t_2]
8. if t_3 < V goto 5
9. j = j - 1 // leader by rule 2, 3
10. t_4 = 4 * j
11. t_5 = a[t_4]
12. if t_5 > V goto 9
13. if i >= j goto 23 // leader by rule 3
14. t6 = 4 * i // leader by rule 3
15. x = a[t6]
16. t7 = 4 * i
17. t8 = 4 * j
18. t9 = a[t8]
19. a[t7] = t9
20. t10 = 4 * j
21. a[t10] = x
22. goto 5
23. t11 = 4 * i // leader by rule 2, 3
24. x = a[t11]
25. t11 = 4 * i
26. t13 = 4 * n
27. t14 = a[t13]
28. a[t12] = t14
29. t15 = 4 * a
30. a[t_15] = x

```

So, the basic blocks are:

1-4 5-8 9-12 13 14-22 23-30

Transformation on basic block

- # A basic block constitutes a set of expressions. There are expressions whose variables are *live* on exit from the block.
- # The calculation of *live variables* is a matter for global flow graph. For now, we are only concerned with local transformations, and we will assume that *live variables* are known before hand.
- # Two basic blocks are said to be *equivalent* if they compute the same set of expressions.

Structure preserving transformations

a. Common subexpression elimination

Consider the basic block:

$$\begin{array}{lll}
 a = b + c & a = b + c & a = b + c \\
 b = a - d & \Rightarrow b = b + c - d & \Rightarrow b = a - d \\
 c = b + c & c = b + c - d + c & c = b + c \\
 d = a - d & d = b + c - d & d = b
 \end{array}$$

b. Dead code elimination

Suppose x is dead, i.e. never subsequently used at the point where the statement $x = y(z)$ appears in a basic block.

c. Renaming temporary variables

Example:

$$t = b + c$$

$$u = b + c$$

We can replace t with u .

- # Suppose we have a statement $t = b + c$ where t is a temporary.
- # If we change this statement to $u = b + c$ where u is a new temporary and change all uses of this instances of t to u . Then, the value of basic block is not changed.

d. Interchange statements

- # Suppose we have two adjacent statements $t_1 = b + c$ and $t_2 = x + y$.
- # We can interchange the two statements without affecting the value of the block if and only if neither x nor y is t_1 and neither b or c is t_2 .

Algebraic transformation

- a. Elimination of statements like $x = x + 0$ and $x = x \times 1$.
- b. Replacement of statements like $x = y^2$ by $x = y * y$.

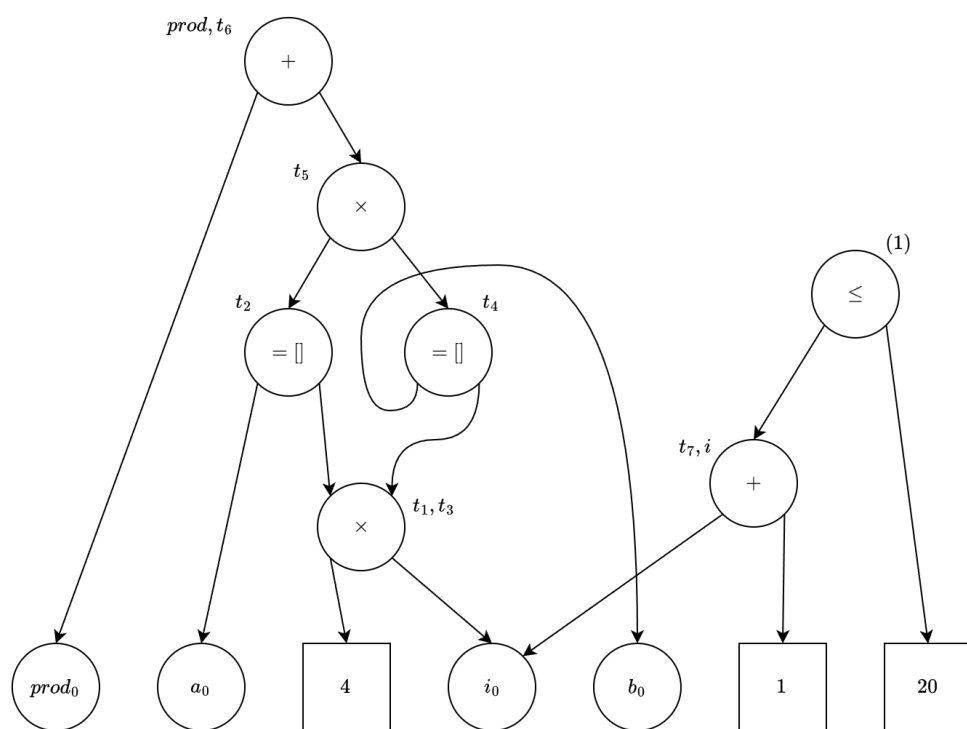
Directed Acyclic Graph (DAG)

- # Constructing a DAG from a three-address is a good way of determination of:
 - a. common subexpression within a block.
 - b. which names are used inside the block but evaluated outside the block.
 - c. which statements of the block could have their computed value used outside the block.
- # A DAG for a basic block is a directed acyclic graph with the following labels on nodes:
 - a. Leaves are labeled by unique identifiers, either variable names or constants. From the operator applied to a name, we determine whether the l-value or r-value is needed. Most leaves represent r-value. The leaves represent initial values of names and we subscript them with zero.
 - b. Interior nodes are labelled by an operator symbol.
 - c. Nodes are also optimally given a sequence of identifiers for labels.

```

1. t1 = 4 * i
2. t2 = a[t1]
3. t3 = 4 * i
4. t4 = b[t3]
5. t5 = t2 * t4
6. t6 = prod + t5
7. prod = t6
8. t7 = i + 1
9. i = t7
10. if i <= 20 goto 1

```



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